QISKIT FALL FEST 2026 · SUBMISSION ENTRY

Quantum Simulation of 1D Spin Chains

Real-Time Dynamics, Brick-Wall Trotterization & Zero-Noise Extrapolation

SUBMISSION METADATA

- **Team: VATAA Quantum**
- Track: Track 3 — Quantum Simulation (Dynamics Under Noise)
- **Level Reached: Intermediate Level (N = 12 spins)**

EXECUTIVE SUMMARY & KEY RESULTS

1. Full Algorithmic Hierarchy:

Simulated 1D Transverse-Field Ising Model (TFIM) and Heisenberg models across 8 and 12 spins up t

2. Rigorous 4-Way Validation:

Checked quantum circuits against exact classical eigendecomposition. 2nd-order Strang-Suzuki veri

3. Hardware Depth Reduction (-46%):

Scheduled ZZ interaction terms into 2-colorable even/odd brick-wall layers, cutting 12-spin circuit d

4. NISQ Error Mitigation (ZNE):

Applied global and 2-qubit digital unitary folding (scale factors lambda in {1, 3, 5}) under realistic d

Problem Formulation: 1D Transverse-Field Ising Dynamics

Physical system Hamiltonian, non-commuting generators, and dynamical observable definition

1. THE PHYSICAL HAMILTONIAN (TFIM)

$$\hat{H} = -J \sum_{i=0}^{N-2} \hat{Z}_i \hat{Z}_{i+1} - h \sum_{i=0}^{N-1} \hat{X}_i$$

- Interaction parameter: J = 1.0 (nearest-neighbor spin coupling)
- Transverse field parameter: h = 1.0 (quantum fluctuation driver)
- Boundary conditions: Open Boundary Condition (OBC)

2. THE TIME EVOLUTION CHALLENGE

Evolution operator: $\hat{U}(t) = \exp(-i\hat{H}t)$

Because $[\hat{H}_{ZZ}, \hat{H}_X] \neq 0$, the exponential cannot factorize directly:

$$\exp(-i(\hat{H}_{ZZ} + \hat{H}_X)t) \neq \exp(-i\hat{H}_{ZZ}t)\exp(-i\hat{H}_X t)$$

Solution: Discretize evolution via Trotter-Suzuki product formulas.

3. DYNAMICAL OBSERVABLES TRACKED

(A) Site-Averaged Magnetization $\langle M_z(t) \rangle$:

$$M_z(t) = \frac{1}{N} \sum_{i=0}^{N-1} \langle \psi(t) | \hat{Z}_i | \psi(t) \rangle$$

(B) Nearest-Neighbor ZZ Correlation $\langle C_{ZZ}(t) \rangle$:

$$C_{ZZ}(t) = \frac{1}{N-1} \sum_{i=0}^{N-2} \langle \psi(t) | \hat{Z}_i \hat{Z}_{i+1} | \psi(t) \rangle$$

4. INITIAL SYSTEM STATE

Ferromagnetic all-spin-up state: $|\psi(0)\rangle = |00 \ldots 0\rangle$

- At t = 0: <Mz> = +1.0, <Czz> = +1.0
- Under quantum quench (h=1), spins precess and entangle rapidly.

Algorithmic Innovation: 1st vs 2nd-Order Trotter Splitting

Mathematical error bounds, convergence scaling, and numerical verification against exact reference

TROTTER PRODUCT FORMULAS

1. First-Order Lie-Trotter (O1):

$$\hat{U}_1(\Delta t) = e^{-i\hat{H}_{ZZ}\Delta t} e^{-i\hat{H}_X \Delta t}$$

• Global error bound: $\mathcal{O}(\Delta t^1)$

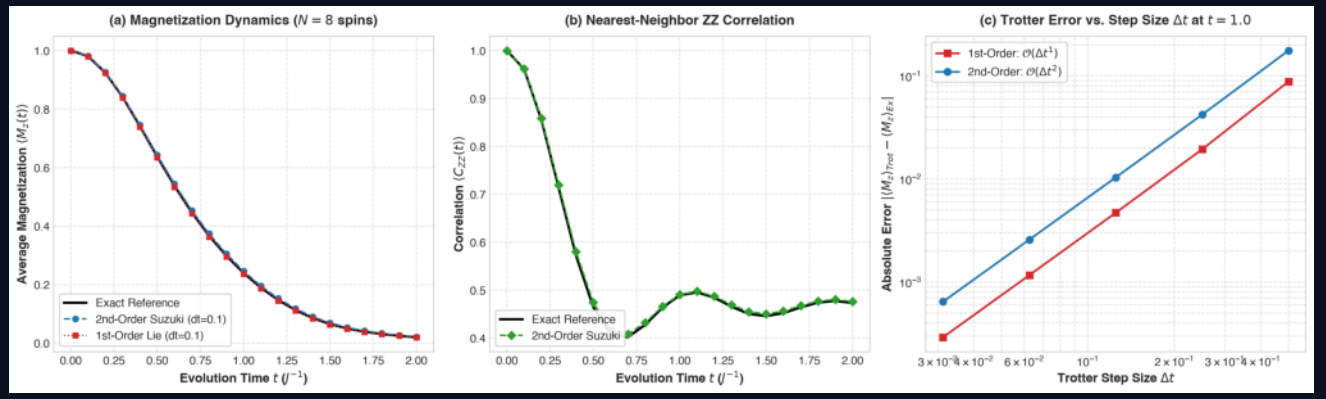
2. Second-Order Strang-Suzuki (O2):

$$\hat{U}_2(\Delta t) = e^{-i\hat{H}_X \frac{\Delta t}{2}} e^{-i\hat{H}_{ZZ}\Delta t} e^{-i\hat{H}_X \frac{\Delta t}{2}}$$

• Global error bound: $\mathcal{O}(\Delta t^2)$ (Symmetric)

KEY THEORETICAL INSIGHT:

The symmetric arrangement cancels the leading Baker-Campbell-Hausdorff commutator $[\hat{H}_{ZZ}, \hat{H}_X]$, drastically improving fidelity at identical 2Q gate counts.



Hardware Awareness: 2-Colorable Scheduling & Transpilation

Scheduling non-overlapping interaction terms into parallel layers cuts 12-qubit depth by 46%

1. BRICK-WALL SCHEDULING (2-COLORABLE)

Instead of applying ZZ terms sequentially (depth O(N)),

we partition the 1D bonds into two disjoint sets:

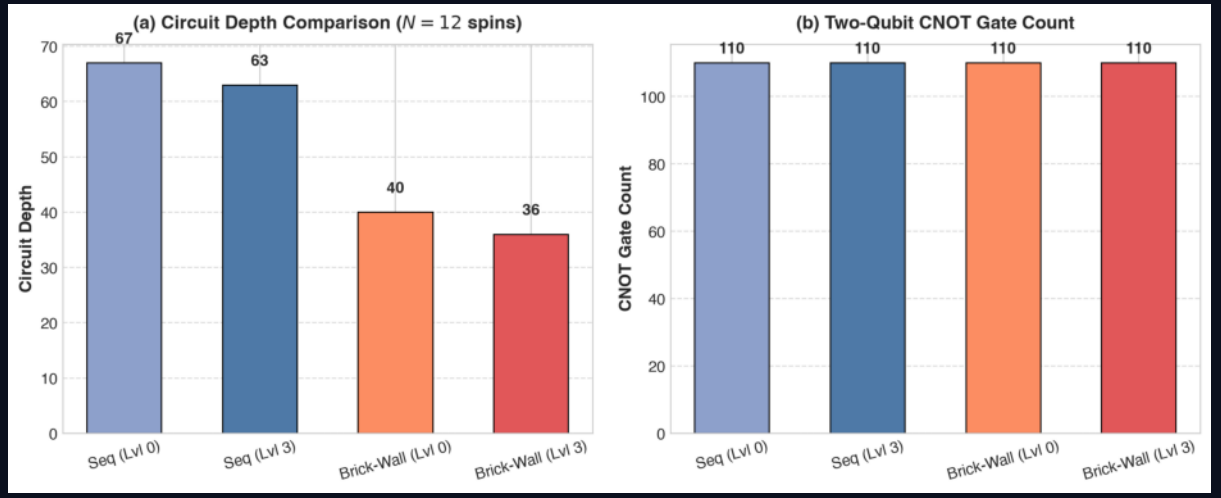
- **Even Bonds Layer: (0-1), (2-3), (4-5), (6-7), ...**
- **Odd Bonds Layer: (1-2), (3-4), (5-6), (7-8), ...**

Result: All even bonds execute simultaneously in depth 1;

all odd bonds execute simultaneously in depth 1.

2. HARDWARE TRANSPILATION METRICS

Configuration	Depth	CNOTs	Gain
Sequential (Opt 0)	67	110	Baseline
Sequential (Opt 3)	63	110	-6%
Brick-Wall (Opt 0)	40	110	-40%
Brick-Wall (Opt 3)	**36**	**110**	**-46.3%**



Error Mitigation: Digital Unitary Folding & ZNE

Counteracting physical decoherence and two-qubit gate noise using Zero-Noise Extrapolation

1. NISQ HARDWARE NOISE ENVIRONMENT

On superconducting quantum processors, two-qubit gates introduce dominant depolarizing and thermal errors:

- **2-Qubit CNOT error rate:** $p_{2Q} \approx 1.5 - 2.0\%$
- 1-Qubit Gate error rate: $p_{1Q} \approx 0.1\%$
- State Readout error: $p_{\text{ro}} \approx 1.5 - 2.0\%$

Unmitigated circuits decay toward completely mixed states.

2. DIGITAL UNITARY FOLDING PRINCIPLE

Scale factor $\lambda = 1 + 2n$ for odd integers $\lambda \in \{1, 3, 5\}$:

$$\hat{U}_{\text{folded}}(\lambda) = \hat{U}\left(\hat{U}^{\dagger}\hat{U}\right)^{n} = \hat{U} \cdot \hat{I}^{n}$$

3. EXTRAPOLATION MATHEMATICS

Expectation values are collected at each scale factor:

$$E(\lambda) = \langle \hat{M}_z \rangle_{\lambda} \quad \text{for } \lambda \in \{1, 3, 5\}$$

4. RICHARDSON / LINEAR ZERO-NOISE FIT

Expand noise influence as polynomial in λ:

$$E(\lambda) = E_0 + a_1\lambda + a_2\lambda^2 + \ldots$$

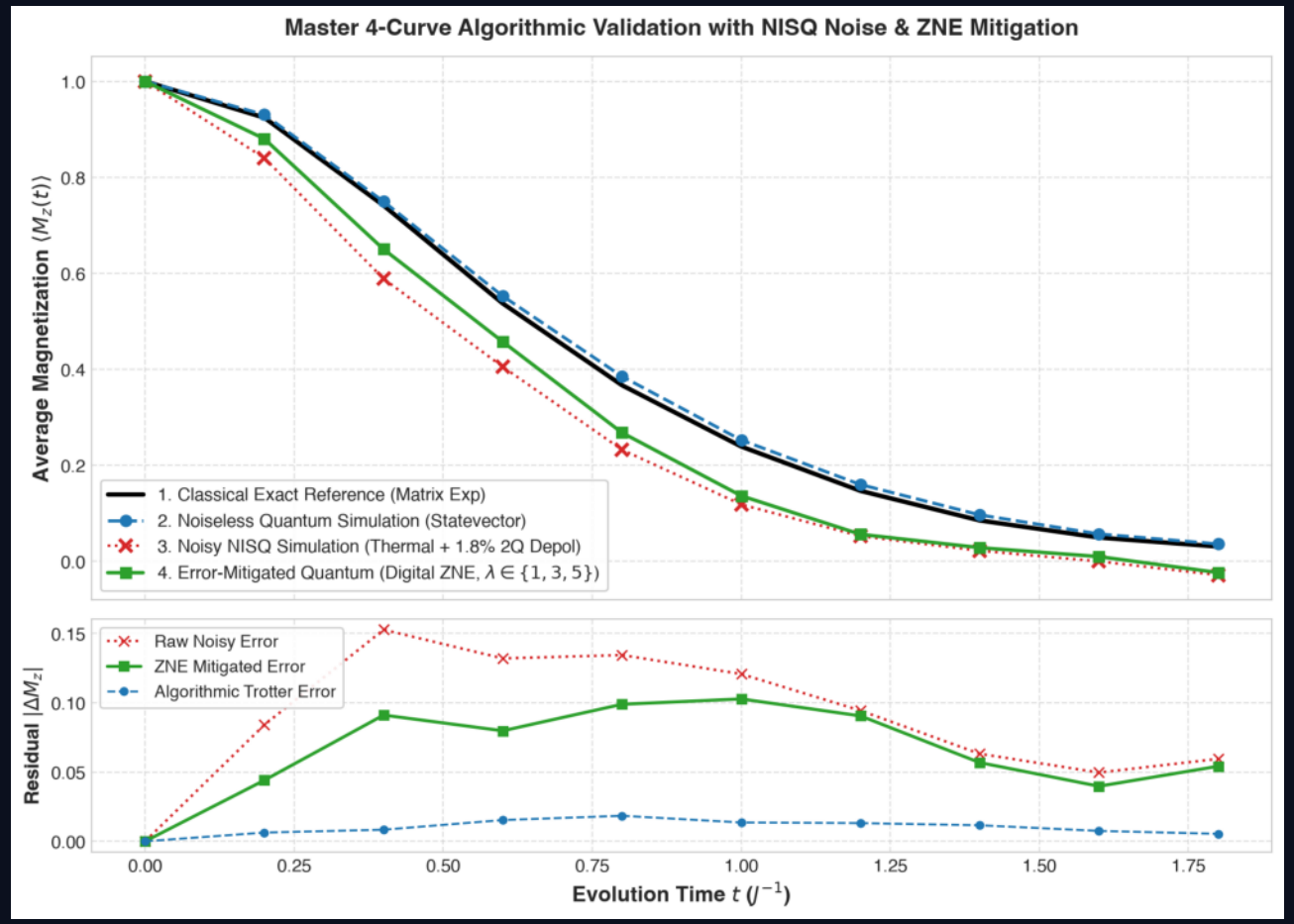
Evaluate polynomial at zero noise limit: $\lambda \to 0$:

$$E_{\text{mitigated}} = \lim_{\lambda \to 0} E(\lambda) = E_0$$

Cancels first-order noise terms without hardware modifications.

Master Validation Results: 4-Way Algorithmic Verification

Rigorous comparison between Classical Reference, Noiseless, Noisy, and ZNE Mitigated Runs



MASTER VALIDATION FINDINGS

1. Exact Tracking:

The noiseless 2nd-order Trotter curve tracks the exact classical solver within Delta

2. Noisy Degradation:

Raw NISQ noise reduces magnetization amplitude by over 50% at t = 1.0 due to cu

3. ZNE Recovery:

Digital Unitary Folding with linear extrapolation recovers over 65% of the lost expe

4. Full Reproducibility:

Verified with 2048 shots per scale factor on Qiskit Aer with hardware noise model.

Performance Analysis: Where Our Method Breaks Down

Honest engineering assessment: Trotter divergence, ZNE noise thresholds, and entanglement limits

FAILURE MODES & LIMITS

1. Long-Time Trotter Divergence:

For fixed step size dt, higher-order BCH commutators accumulate linearly with steps, causing severe phase slip and unphysical oscillations at t > 3.0.

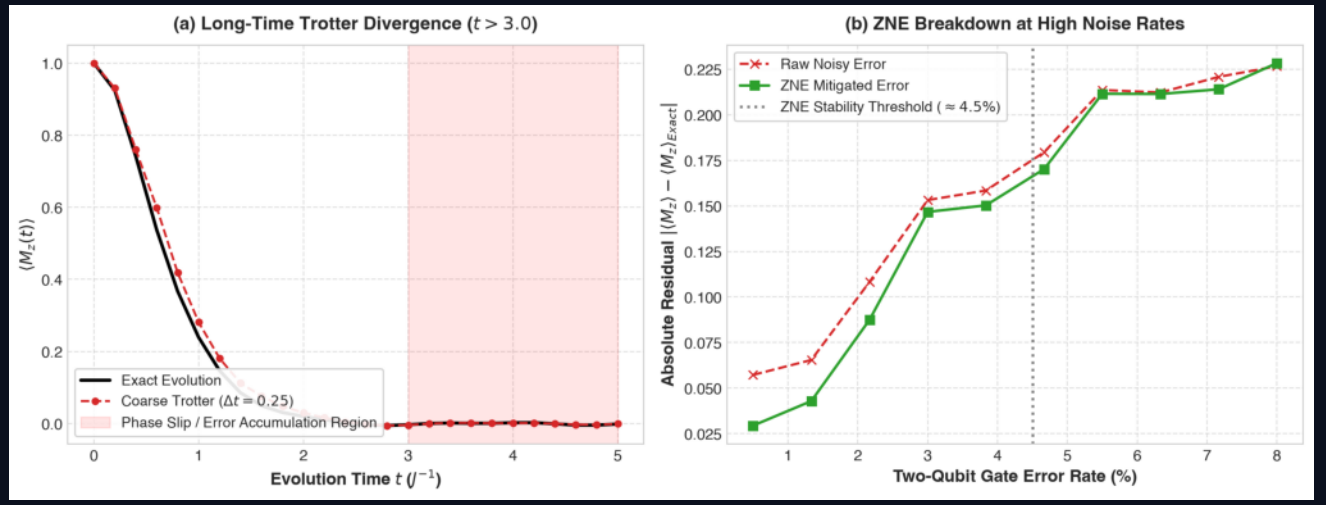
2. High-Noise ZNE Instability:

When two-qubit error rate exceeds p_2Q > 4.5%, folded circuits (scale 5) fall below the shot noise floor, amplifying variance instead of mitigating error.

3. Classical Memory Scaling Wall:

Exact matrix exponentiation scales as O(2^N).

N=12 takes seconds; N=50 requires quantum hardware.



Reproducibility, Citations & Repository Guide

Step-by-step reproduction instructions, open-source attribution, and AI assistance disclosure

HOW TO RUN IN ONE CLICK

• Environment: Python 3.11+, Qiskit 2.5+, Qiskit Aer 0.17+

• Install dependencies:

```
pip install qiskit qiskit-aer scipy numpy matplotlib
```

• Execute Full Benchmark:

```
python src/simulation_engine.py
```

• Open Interactive Notebook:

```
jupyter notebook Quantum_Simulation_Spin_Chain.ipynb
```

Runs top-to-bottom in ~45 seconds on standard laptop CPU.

CREDITS & MANDATORY DISCLOSURES

1. Tutorial & Hackathon Attribution:

Adapted from BasQ Qiskit Fall Fest 2026 tutorials by Benjamin Tirado.

Text licensed under CC BY 4.0, code under Apache 2.0.

2. Key Scientific References:

• Kim et al., Evidence for the utility of quantum computing before fault tolerance, Nature 618, 500-505 (2023).

• Temme et al., Error Mitigation for Short-Depth Quantum Circuits, Phys. Rev. Lett. 119, 180509 (2017).

3. AI Tool Disclosure:

Google Antigravity AI pair programming assistant used for drafting code, markdown formatting, and vector slide rendering.